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VIGENÈRE CIPHER



The Vigenère cipher, misattributed to 19th-century cryptographer Blaise de Vigenère (others had independently invented it earlier), was impossible to crack for hundreds of years. It is essentially the Caesar cipher, except it makes use of a multipart key. The so-called *Vigenère key* is a word, or even a random series of letters. Each letter represents a number by which to shift the letter in the message: *A* represents shifting a letter in the message by 0, *B* represents 1, *C* represents 2, and so on.

For example, if a Vigenère key is the word “CAT,” the *C* represents a shift of 2, the *A* represents 0, and the *T* represents 19. The first letter of the message gets shifted by 2, the second letter by 0, and the third letter by 19. For the fourth letter, we repeat the key of 2.

This use of multiple Caesar cipher keys is what gives the Vigenère cipher its strength. The possible number of combinations is too big to brute force.